

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

SA3 – General Insurance

May 2025

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q.1)

- i) When a new general insurance company begins operations, it must comply with several regulatory requirements, particularly set by the Insurance Regulatory and Development Authority of India (IRDAI). These are designed to ensure financial stability, policyholder protection, and market discipline.

Capital Requirements

- The investors/promoters are subject to lock in requirements as per regulations

Solvency Margin

- Insurers must maintain a solvency ratio of at least 1.5 times the required solvency margin.
- This ensures that the company has enough capital to meet its liabilities even in stress scenarios.

Investment Regulations

- IRDAI prescribes strict investment norms to ensure safety and diversification:
 - A specific percentage must be invested in government securities.
 - Caps on exposure to equity, real estate, or group companies.
 - Mandatory use of Approved Investment lists.

Product Approval

- New products should be filed and approved as per the Insurance Product Regulations
- Pricing must be objective and actuarially justified and may be subject to regulatory review

Product Approval

The company would need to onboard all the required KMPs before the start of business.

Reinsurance Requirements

- Insurers are required to cede a portion of their risk to reinsurers.
- A portion must be offered to General Insurance Corporation (GIC Re), the national reinsurer, under compulsory cession rules.
- Additionally, the company should have a board approved Reinsurance policy

Operational Guidelines

- Need for IRDAI approval on:
 - Opening branches
 - Appointing key management personnel (KMPs)
 - Change of control, mergers, or acquisitions

Distribution and Commission Limits

- Restrictions on:
 - Commission to agents and intermediaries
 - Use of corporate agents, brokers, web aggregators
- Mandatory registration and licensing of intermediaries.

Policyholder Protection & Grievance Redressal

- Must establish a policyholder grievance cell.

- Disclosures on products, claim procedures, turnaround time, etc., are mandatory.

Statutory Reporting & Audits

- Regular submission of:
 - Financial statements
 - Actuarial reports
 - Solvency and investment compliance reports
- Subject to IRDAI inspections and audits.

Technology and Cybersecurity Guidelines

- Compliance with IRDAI's norms on IT systems, cybersecurity, and data privacy, especially if selling products online or using digital platforms.
- The company would need financial and IT systems in place before the start of operations

Policies:

It will be mandatory to have board approved policies and committees with respect to the following:

- Underwriting Policy
- Claims Policy
- Reinsurance policy
- Outsourcing policy
- Claims committees
- Grievance committee
- Outsourcing committee etc

(Max 10)

- ii)** Risk measure is TVaR 99% as it considers tail risk associated in 1% worst outcomes.

$$\text{TVaR}_{99\%}(\text{Health}) = ((-50\%) + (-50\%) + (-50\%) + (-50\%) + (-50\%) + (-100\%) + (-100\%) + (-100\%) + (-100\%) + (-100\%))/10 = -75\%$$

$$\text{TVaR}_{99\%}(\text{Crop}) = ((0\%) + (0\%) + (0\%) + (0\%) + (0\%) + (0\%) + (-100\%) + (-100\%) + (-200\%) + (-200\%))/10 = -60\%$$

Allocate $-75\%/(-75\%-60\%) = 56\%$ to health LoB and

$-60\%/(-75\%-60\%) = 44\%$ to Crop lob

$$\text{RAROC}_{\text{health}} = 15\%/56\% = 27\%$$

$$\text{RAROC}_{\text{crop}} = 20\%/44\% = 45\%$$

Crop will make a more efficient use of capital since its RAROC is higher than that of Health LoB

(Max 5)

iii)

- Diversification – Health and Crop are not perfectly correlated, so the insurer may achieve better expected returns on capital by writing Health and Crop and diversifying its portfolio.
- Model Risk – It is possible that the simulation projections rely on an incorrect model, and Crop does not always provide a significantly better risk-adjusted return on capital than Health.
- Volume Limitations – Since premium was not given for each line, it is possible that there are limits to the amount of growth that management can achieve in Crop given the government tender associated with it. Thus, as long as Health also satisfies management's cost of capital, it may make sense to also write Health.

- Customer Needs – It may be necessary to write Health to support Crop to satisfy the customer's needs and also maintaining a brand name with the general public as Crop being a niche line may not be necessary be popular among the general public.

(Max 4)

iv)

- Loss Ratio- Helps assess the adequacy of premiums relative to claims. A high loss ratio may indicate that premiums are insufficient.
- Expense Ratio- Evaluates the efficiency of operations. A lower expense ratio indicates better cost management.
- Combined Ratio- A combined ratio below 100% indicates profitability, while above 100% indicates a loss.
- Return on Equity (ROE)- Measures the profitability relative to shareholders' investments
- Premium Growth Rate- Indicates market acceptance and growth potential of the company's products.
- Market Share-Indicates competitive positioning and growth potential in the market.
- Solvency Margin- Essential for assessing financial health and regulatory compliance.
- Claims Frequency and Severity- Understanding these metrics helps assess the risk associated with each line of business.

(Max 5)

v)

Call the combination of crop and health lines as Portfolio D

$$E(r_D) = 64.8\% \cdot 15\% + 35.2\% \cdot 20\% = \sim 16.8\%$$

$$\text{Var}(D) = (64.8\% \cdot 20\%)^2 + (35.2\% \cdot 33\%)^2 + 2 \cdot (64.8\% \cdot 35.2\% \cdot 5\%) = \sim 5.31\%$$

	Health and Crop	Government security	Equities
Expected return	16.8%	7%	12%
S.D	23%	0%	30%

The insurance portfolio represents underwriting profit/loss, relatively high return with moderate to high risk.

The government security is risk-free investment, guarantees a stable return with no volatility.

The equity investment represents high volatility, return is lower than insurance portfolio but higher than government security.

We can use the Sharpe Ratio to assess efficiency of return relative to risk. Assuming a risk-free rate of 7% (from government securities):

1. Insurance Portfolio (Health + Crop)

$$\text{Sharpe Ratio} = (16.8\% - 7\%) / 23\% \approx 43\%$$

2. Equity

$$\text{Sharpe Ratio} = (12\% - 7\%) / 30\% \approx 16.7\%$$

3. Government Security

$$\text{Sharpe Ratio} = (7\% - 7\%) / 0\% = 0(\text{undefined, treated as } 0)$$

- The insurance business portfolio offers the highest risk-adjusted return.
- Equity, while offering decent returns, are not as efficient per unit of risk as underwriting business.

- Government securities offer safety and capital preservation but are not growth-oriented.

It may be considered to combine the insurance operations with a small allocation to equity to enhance diversification without significantly increasing overall portfolio risk, as long as correlation assumptions are favorable.

(Max 8)

vi)

Basic Structure of Catastrophe Models:

Catastrophe models are designed to estimate potential losses from rare but severe events (like floods, droughts, earthquakes, etc.). They typically have five core components:

1. Event module
 - Database of historic events affecting crop defined by its physical parameters, location and annual probability/frequency of occurrence.
2. Hazard Module
 - Simulates the physical characteristics of a catastrophic event (e.g., rainfall, windspeed, drought intensity).
 - Includes frequency, location, severity.
3. Exposure Module
 - Details the assets or insured entities exposed to the hazard.
 - In crop insurance, this includes information on crop types, acreage, geography, and planting schedules.
4. Vulnerability Module
 - Translates hazard intensity into expected damage or loss.
 - Considers crop susceptibility to drought, flood, or other stress.
5. Financial Module
 - Converts physical losses into monetary loss for the insurer.
 - Applies policy terms like deductibles, limits, and reinsurance.

(Max 5)

vii)

- Geographical Diversification-
 - Spread the portfolio across regions with low correlation of climate risks.
 - Reduces aggregate exposure to a single localized catastrophe.
- Use of Weather Derivatives or Parametric Insurance
 - Pay based on predefined triggers (e.g., rainfall below a threshold) instead of actual loss.
 - Allows faster claims and reduces model uncertainty.
- Improved Data & Satellite Monitoring
 - Use remote sensing and real-time agrometeorological data for better exposure and vulnerability assessments.
- Reinsurance & Risk Transfer
 - Transfer peak catastrophe risk to reinsurers or capital markets (e.g., cat bonds).
 - Helps stabilize insurer solvency in extreme years.
- Better Farming Practices & Education
 - Promote drought-resistant crops or climate-resilient techniques among farmers.
 - Indirectly reduces loss severity.

(Max 5)

viii) Liquidity Requirements

- The insurer must maintain sufficient liquid assets to cover claims during peak payout seasons (especially for crop insurance, often post-harvest).

- Quick convertibility into cash is key during natural disaster events.

Nature and Duration of Liabilities

- Health Insurance: Claims are generally short- to medium-term in nature, with frequent but relatively predictable cash outflows.
- Crop Insurance: Highly seasonal and catastrophe-prone, with potential for large, concentrated payouts due to weather events.

Investments should be relatively short- to medium-duration, with high liquidity to match claim patterns.

Risk Appetite and Volatility Tolerance

- Crop claims are influenced by climatic and catastrophe risks, making this portfolio more volatile.
- Investment policy should avoid high-risk assets that could worsen financial strain during large claim events.

Conservative investment strategies are preferred to preserve capital during adverse periods.

IRDAI Guidelines

- Required to invest a portion in government securities and other approved safe instruments.
- Limits on investments in equities, group companies, or real estate.

Asset-Liability Matching (ALM)

- Align investment maturities with expected timing of claims (e.g., short bonds for health, seasonal alignment for crop).
- Inflation protection might be considered, as health costs and crop losses are sensitive to inflation and commodity prices.

Diversification and Correlation

- Avoid high correlation between investment risks and underwriting risks.
 - E.g., investing heavily in agri-sector Equities while underwriting crop insurance increases risk concentration.

Catastrophe and Reinsurance Strategy

- Since crop is prone to catastrophic losses (floods, droughts), reinsurance plays a key role.
- Investment policy should complement the reinsurance program, ensuring funds are available for large retentions or delayed recoveries.

Macroeconomic Factors

- Monitor monsoon forecasts, inflation trends, and health sector developments, which can impact both claims and investment returns.
- Policy should be flexible enough to adjust asset allocation as these risks evolve.

An insurer must adopt a conservative, liquid, and diversified investment policy that ensures stability, solvency, and regulatory compliance, while remaining agile to meet the unique seasonal and catastrophic risks of its portfolios.

(Max 9)
[51 Marks]

Q.2)

- i) The industrial property line is characterized by:

- Large deductibles
- High-severity, low-frequency claims
- Quick claim reporting, but
- Volatile reserve development, due to inaccurate initial reserve estimates

Concerns with the Chain Ladder Method

- Assumes historical development patterns will continue into the future
- Sensitive to data volatility, especially when there are few large claims
- Heavily dependent on early incurred amounts, which may be inaccurate or volatile in this line due to potential case reserving inconsistencies

Thus, the chain ladder has produced unstable reserve estimates, often over- or under-projecting due to outliers or immature data.

(Max 5)

ii) The BF method blends:

1. Expected Loss Ratio (ELR) based projections, and
2. Actual loss development to date

This allows to:

- Rely less on possibly volatile early data
- Incorporate a priori assumptions (e.g., pricing or underwriter estimates)
- Smooths out large outlier claims
- Provide greater stability in reserve estimates

(Max 3)

iii) Recommendation

Given the characteristics of our industrial property line, the Bornhuetter-Ferguson method offers a more robust and stable approach to reserve estimation, particularly during early development.

- Transitioning to BF for recent accident years
- Using a hybrid approach: BF for immature years, Chain Ladder for more developed years
- Regularly reviewing and updating expected loss ratios based on pricing, exposure trends, and experience

Next Steps

1. Set ELR assumptions per underwriting year or segment
2. Run side-by-side comparisons (BF vs Chain Ladder) for recent years
3. Discuss revised reserve estimates with finance and claims teams
4. Document methodology for audit and regulatory purposes

(Max 4)

iv) Changes in Claims Handling Practices

- Faster or slower settlement practices affect the speed of development.

- Introduction of centralized claims teams, automation, or specialized adjusters may accelerate or delay reporting/reserving.
- New reserving philosophies (e.g., more conservative or aggressive case reserving) can artificially inflate or deflate early incurred values.

Change in Reinsurance Structure

- Introducing or altering retentions, excess-of-loss, or quota share treaties may affect what's reported net of reinsurance.
- Higher deductibles or retentions might delay development until large losses pierce layers.

Changes in Underwriting or Pricing Strategy

- Writing new segments of business or expanding into different geographies may introduce loss patterns not present in past data.
- Stricter underwriting might reduce loss frequency but not severity, altering the curve shape.

Changes in Policy Terms & Conditions

- Introduction of higher deductibles, co-insurance, or coverage exclusions can shift the nature and timing of claims.
- For example, large deductibles in industrial property may delay significant reserve recognition until claims exceed thresholds.

IT or System Changes

- Migration to a new claims system can lead to reporting lags, data quality issues, or inconsistency in development tracking.
- Historical data may not be perfectly mapped or categorized post-migration.

(Max 5)

v) Changes in Claims Handling Practices

Compare cumulative reported and paid over time — sudden changes in slopes can be revealing

- Sudden acceleration or delay in development.
- Earlier or later than usual jumps in reported or paid losses.
- Step changes in link ratios across development years (e.g., Dev 1–2 goes from 1.4 to 1.1 after change).

Change in Reinsurance Structure

Analyse gross vs net triangles side-by-side to detect the shift.

- A drop in large claims being reported or paid net of reinsurance post-change.
- In triangles: lower cumulative values in recent years, especially in later development periods.
- Distorted tail factors if prior large losses are no longer reflected.

Changes in Underwriting or Pricing Strategy

Examine average claim size or claim count triangles (if available).

- Lower claim frequencies but stable or volatile severities.
- Newer accident years develop differently (slower or faster) than prior years.
- Inconsistencies in development factors by accident year.

Changes in Policy Terms & Conditions

Use a triangle of number of claims or frequency rates for early detection.

- Delayed development: e.g., higher deductibles mean claims appear later.

- Development patterns shift toward heavier tail.
- Fewer small claims being reported in early development.

IT or System Changes

Sudden changes in triangle shape that do not align with real-world events are red flags.

- A specific cohort (accident or calendar year) shows irregular development (e.g., flat or missing development).
- Unexpected drops or spikes not tied to actual claim events.
- Data inconsistencies across development years.

(Max 10)

vi) Surplus Reinsurance

- It is Proportional treaty reinsurance where the ceding company retains a fixed amount of risk (called the retention), in this case- INR 500,00,000, and the surplus portion above that is shared with reinsurers.
- Expressed in “lines” (e.g., 4-line surplus means the reinsurer accepts up to 4 times the retention).
- Premium is Shared between insurer and reinsurer in proportion to their share of the risk.
- Claims also shared proportionally.
- Suitable for risks with varying sum insured amounts, like large commercial property portfolios.

Individual Excess of Loss Reinsurance

- It is non-proportional treaty reinsurance where the insurer retains losses up to a predetermined limit (the retention) – in this case INR 200,00,000, and the reinsurer covers the excess up to a maximum limit (the cover) in this case- up to maximum limit of INR 500,00,000.
- Premium is based on the reinsurer’s exposure to loss, often rated using risk modelling.
- Claims only kicks in when individual claims exceed the retention.
- Useful for protecting against large, infrequent claims from a single property (e.g., fire at a warehouse).

Facultative Reinsurance

- It Can be proportional or non-proportional; negotiated individually.
- Reinsurance is purchased for a specific risk or policy. The reinsurer evaluates and decides whether to accept the risk.
- Premium and Terms: Negotiated case-by-case.
- Claims: Paid based on the individually agreed contract terms.
- Usage: Used for large or unusual risks outside the scope of existing treaties (e.g., a high-value factory or one-off complex risk).

(Max 6)

vii) Sum Insured / Policy Limit

- High-value risks may exceed the company’s risk appetite or retention capacity.
- Larger sums insured may require greater cession to manage exposure and capital requirements.

Hazard Profile and Risk Type

- Examples: Industrial vs. residential property; exposure to fire, flood, earthquake, etc.
- Riskier exposures (e.g., chemical plants, warehouses with flammable materials) may need greater reinsurance support.

Location

- Properties in catastrophe-prone areas (earthquakes, hurricanes) may require higher cession.
- Geographic risk concentrations increase potential for catastrophic losses.

Loss History

- Why it matters: A poor claims history may indicate higher future losses.
- Impact: Higher past losses may encourage a company to cede more risk.

Retention Strategy

- The insurer's internal capital position and risk appetite determine how much it retains.
- More conservative companies will cede more to manage volatility.

Premium Adequacy

- If the premium is inadequate for the level of risk, the company may reduce its exposure through cession.
- Helps avoid under-pricing losses.

Availability and Terms of Reinsurance

- Ceding decisions depend on capacity, pricing, exclusions, and the reputation of reinsurer.
- Better terms or expertise from reinsurer may encourage higher cession.

(Max 6)
[39 Marks]

Q. 3)

i)

a) Implementation of wellness and health programs:

- These programs are likely to reduce the frequency and severity of claims, as healthier employees generally lead to fewer and less costly claims.
- Premium impact: The insurer may reflect this positive risk mitigation by applying a credibility-adjusted experience modification, leading to a discount or reduction in premium, assuming sufficient data supports the effectiveness of the programs.

b) Doubling of the number of employees:

- The increase in exposure (more covered lives) means a higher total expected claims cost, even if the per-employee risk remains the same or improves slightly.
- Premium impact: The base premium would increase proportionally to reflect the greater exposure. However, the insurer might also apply volume discounts or economies of scale, which could slightly reduce the per employee premium rate.

(Max 4)

ii) Advantages

- Hours driven correlates more closely with accident likelihood than simple ownership for number of years.
- Drivers who drive more are statistically more likely to be prone to accidents.
- Reduces cross-subsidization (e.g., between low-mileage and high-mileage drivers).
- Reflects actual usage behaviour rather than just vehicle presence.
- Encourages responsible driving habits and reduced usage, which may benefit society (e.g., lower emissions, congestion).
- Aligns well with telematics programs.

- Can enable pay-as-you-drive (PAYD) or pay-how-you-drive (PHYD) models.

Challenges & Limitations

- Requires accurate tracking of hours driven (via telematics, apps, odometers, etc.).
- Raises privacy concerns and potential data accuracy issues.
- Current systems, rating structures, and actuarial models are built on car-years.
- Moving to hours-driven requires significant changes to:
 - Rating algorithms
 - Policy administration systems
 - Reporting structures
- Drivers may find this intrusive or difficult to understand.
- May be perceived as a "surveillance" tool unless well-positioned.
- Early data may be incomplete or skewed.
- Different tracking devices or opt-in rates might lead to selection bias (e.g., only low-risk drivers opting in).
- Policyholders may change driving behaviour over time — making projections less stable.
- Seasonal and lifestyle variations (e.g., vacations, remote work) can complicate reserving and pricing.

(Max 6)

[10 Marks]
